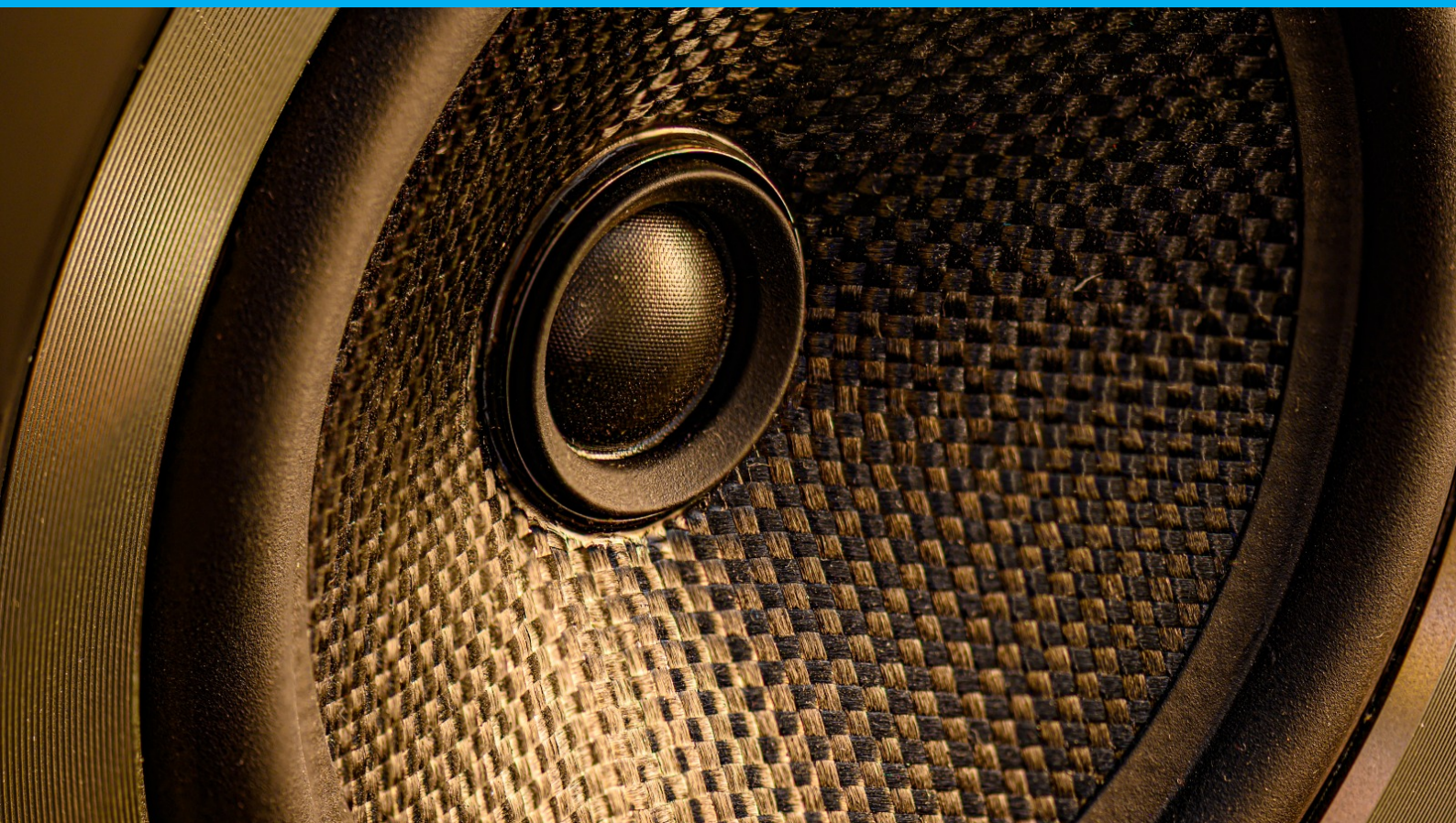


EE1L1 IP-1 Project Report

Power Supply Design

Group B2-1 PS1



EE1L1 IP-1 Project Report

Power Supply Design

by

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EE1L1

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Contents

1	Introduction	2
2	Design Methodology	3
3	Simulation results	4
4	Power Supply Assembly	5
5	Measurement results	6
A	Appendix	8
A.1	Tips	8
A.2	Resources	8
B	Measurements	9
B.1	Ripple Voltage.	9
B.2	Discharge Time Capacitors	12

Guidelines

Remove this file This document intends to provide a generic report structure which your group can use for the intermediate and or final report. Note that your group may need to adjust the template according to the guidelines provided for the intermediate and or final report.

Please note that you are expected to remove the guidelines provided in this template you submit, i.e. Incorporate your original work into this template and remove all the material presented as guidelines here.

Task division:

1. Alex
2. Alex
3. Gerrald
4. Alex
5. Gerrald
6. Together

Merging, handing in and removing instructions: Gerrald

[1] [2]

1

Introduction

The Introduction of the power supply report can contain: 1. The system requirements, the complete schematic of the power supply included. Note: Take care that each element is clearly and unambiguously marked. 2. The problem definition, in the form of the principal choices for reaching the required performance, and a summary of the steps to follow.

-Give a summary of relevant background as well as the objectives of the project. -Include the general requirements/ limitations/ constraints of the project. -You can present the organization of the document at the end of the introduction.

2

Design Methodology

1. Contains a general description of the design procedure. 2. Start by briefly discussing the role of the elements in the circuit schematic of the power supply to be designed. Note: In case of elaborate expressions needing being derived, please consider moving them to appendices, by giving proper references, and keep in this subsection the final results, only. -Note that the theory included in IP-1 manual is not needed to be repeated in this chapter, just refer to the IP-1 manual.

-Describe how you solved the technical problem (which methods or techniques did you use?)

-Give an overview picture/figure/chart of your design methodology helps the reader better understand your approach.

-Give the key design equations.

-Include the calculations and computations (if they are too long, move a part of them to appendices).

3

Simulation results

- Use tables, charts, figures, and graphs to best present your findings.
 - Move too large quantities of raw data, detailed equation derivations, or code to appendices.
 - Simulation results should be critically compared to theory/methodology.
 - Address the limitations/constraints in the theory/methodology.
1. Perform the simulations in LTspice.
 2. Include a properly legible schematic of the simulated circuit.
 3. Some of the obtained plots should be included in the report; pay special attention to ensuring the legibility of these plots. Make a judicious selection of presented results, with one or two essential plots being included in this subsection and all other results being included in appendices, by giving proper references.
 4. Compare simulated and previously obtained analytic results and attempt to explain possible discrepancies
-

Figure 3.1: The simulated circuit

4

Power Supply Assembly

Description of the assembling of the power supply. 1. Explain your choices for the circuit elements and justify possible deviations from the previously designed values. 2. Should redesign be dictated by the unavailability of some components, please include a discussion on how to deal with these unavailabilities.

5

Measurement results

- Use tables, charts, figures, and graphs to best present your findings.
- Move too large quantities of raw data, detailed equation derivations, or code to appendices.
- Measurement results should be critically compared to theory/methodology.
- Address the limitations/constraints in the theory/methodology.

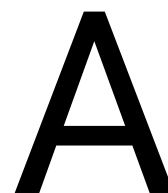
1. Describe the procedure which you followed to perform the measurements. 2. Include illustrating snapshots of waveforms; pay attention to the legibility of the plots and/or images. 3. Compare measured results with simulated and/or analytic results and attempt to explain possible discrepancies.

Some measurements were done to characterize the power supply and measure its performance. The following characteristics were measured:

- The average output voltage under various levels of load.
- The ripple in the output voltage under various levels of load.
- The discharge time of the capacitors after operation.

Bibliography

- [1] Roel Grit. *Project Management: A Practical Approach*. 4th ed. Groningen, The Netherlands: Noordhoff Uitgevers, 2015.
- [2] Dr.ir. G.J.M. Janssen et al. *Integrated Project 1 (IP-1) EE1L1 "Booming Bass (Sound) System"*. Delft University of Technology. 2024.



Appendix

This chapter presents additional tips and resources.

A.1. Tips

The students should prepare the reports in a proper format. This means that they should use commonly accepted font and font size, page margins, heading sizes etc. Graphs and tables should be legible on paper (after printing), should have captions in a matching style, etc. They should not forget to include cover, title page, table of contents for the report.

A suggested template is provided on overleaf <https://www.overleaf.com/read/xdynzwbcdvzy>

A.2. Resources

You may also use the $\LaTeX$ templates provided in the sources listed below:

- TU Delft corporate identity page provides additional resources for $\LaTeX$: www.tudelft.nl/huisstijl/downloads
- Overleaf also presents additional template options: www.overleaf.com/edu/tudelft#templates

B

Measurements

B.1. Ripple Voltage

The voltage waveform over a resistor was measured four times under four conditions:

- With the output connected to two $38\ \Omega$ resistors in series, forming a $19\ \Omega$ resistor.
- With the output connected to a $38\ \Omega$ resistor.
- With the output connected to two $38\ \Omega$ resistors in parallel, forming a $76\ \Omega$ resistor.
- With no load on the output.

The graphs can be seen in figures B.1, B.2, B.3, and B.4. When conducting those measurements, only the positive side of the power supply was loaded, because the negative side is an exact copy of the positive one.

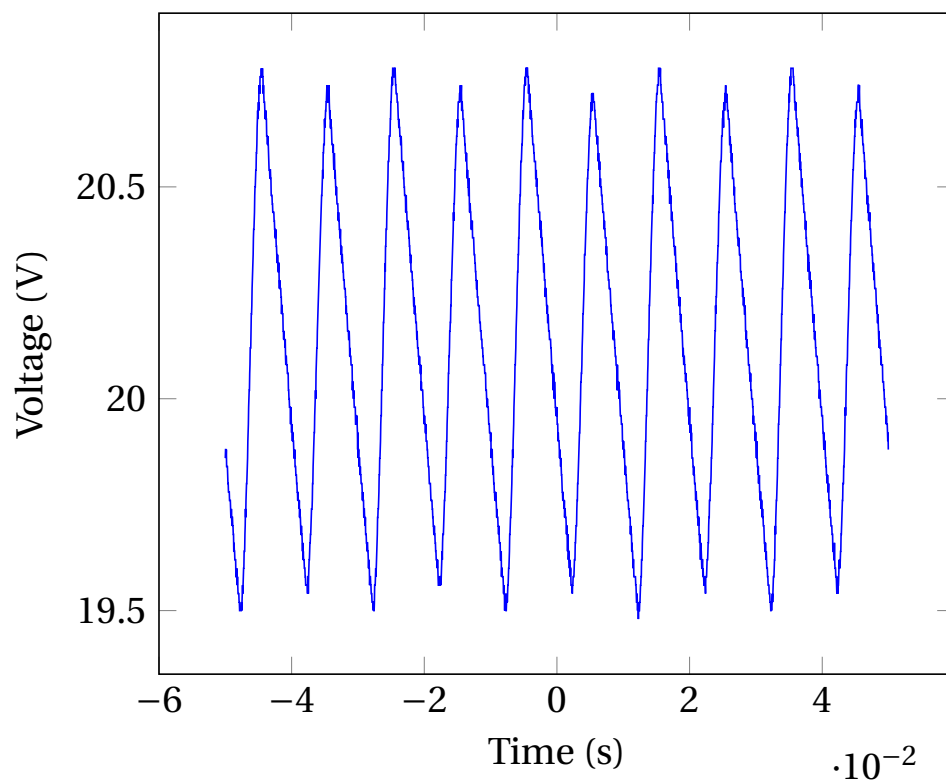


Figure B.1: The output voltage waveform with the power supply loaded with a $19\ \Omega$ resistor.

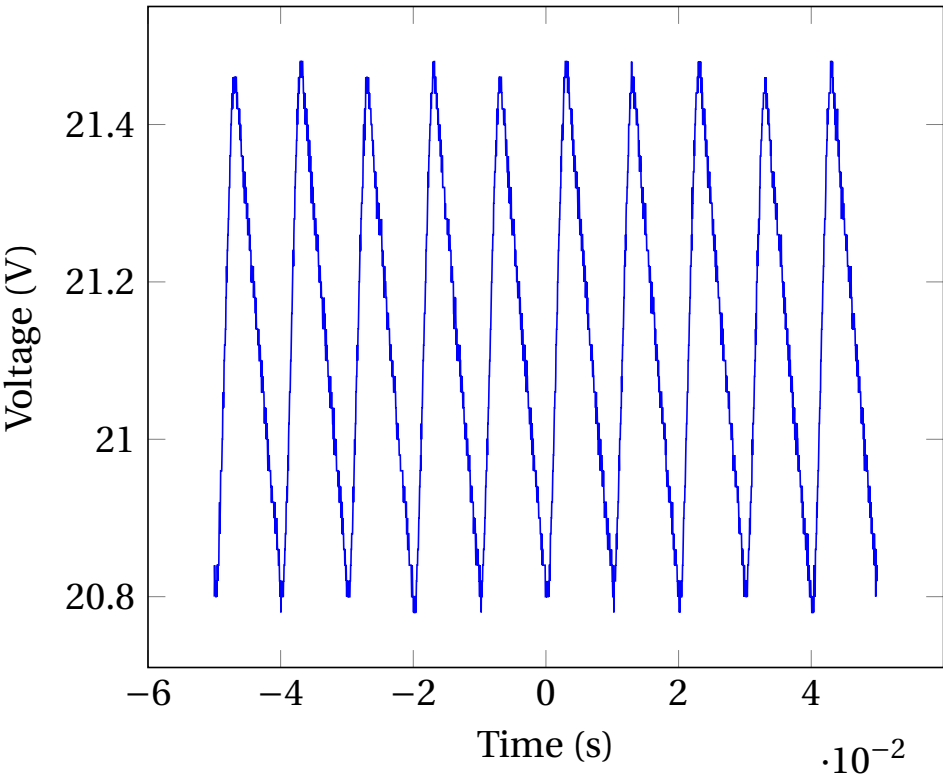


Figure B.2: The output voltage waveform with the power supply loaded with a 38 Ω resistor.

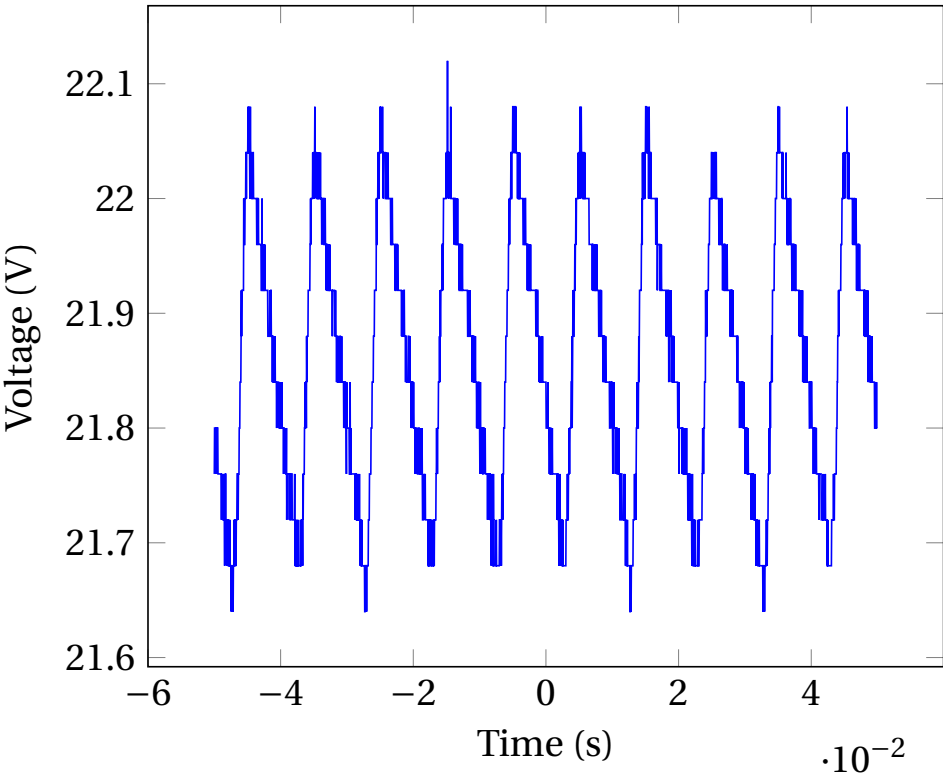


Figure B.3: The output voltage waveform with the power supply loaded with a 76 Ω resistor.

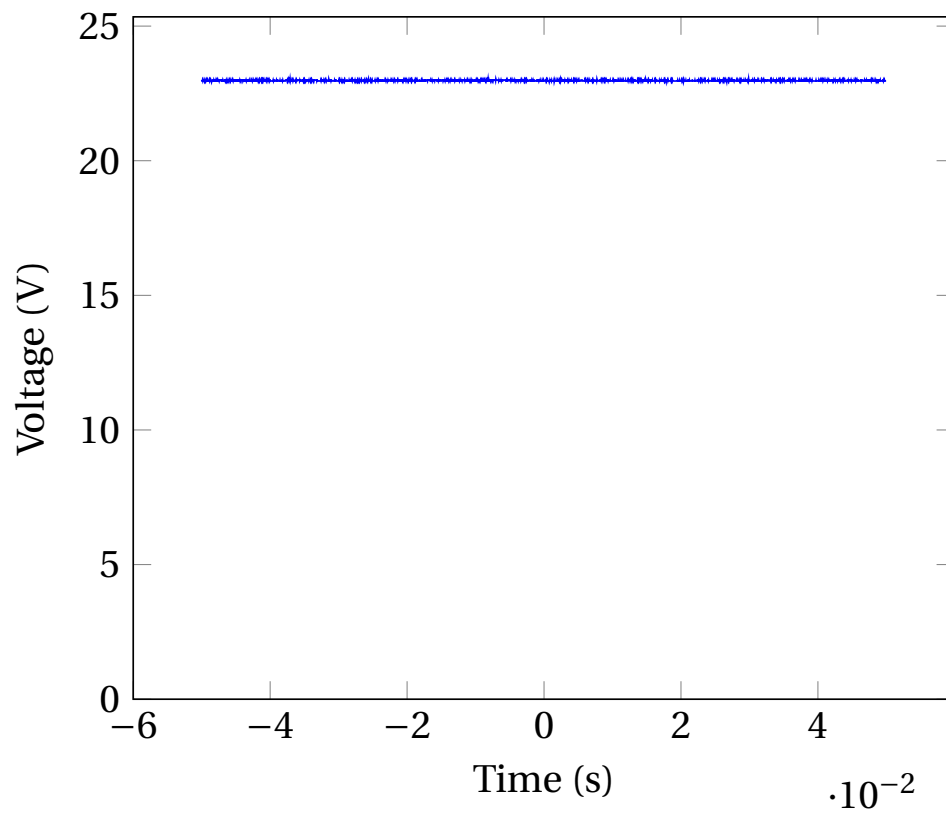


Figure B.4: The output voltage waveform with the power supply under no load.

Table B.1: Characteristics of the power supply under various levels of load

Resistance Load (Ω)	Average Voltage (V)	Average Current (A)	Ripple (%)
19			
38			
76			
Open	23	0	

Those measurements led to the following characteristics which can be seen in table B.1. The ripple was calculated using the following formula [2]:

$$U_{ripple}[\%] = \frac{u_{max} - u_{average}}{u_{average}} \cdot 100\%$$

B.2. Discharge Time Capacitors

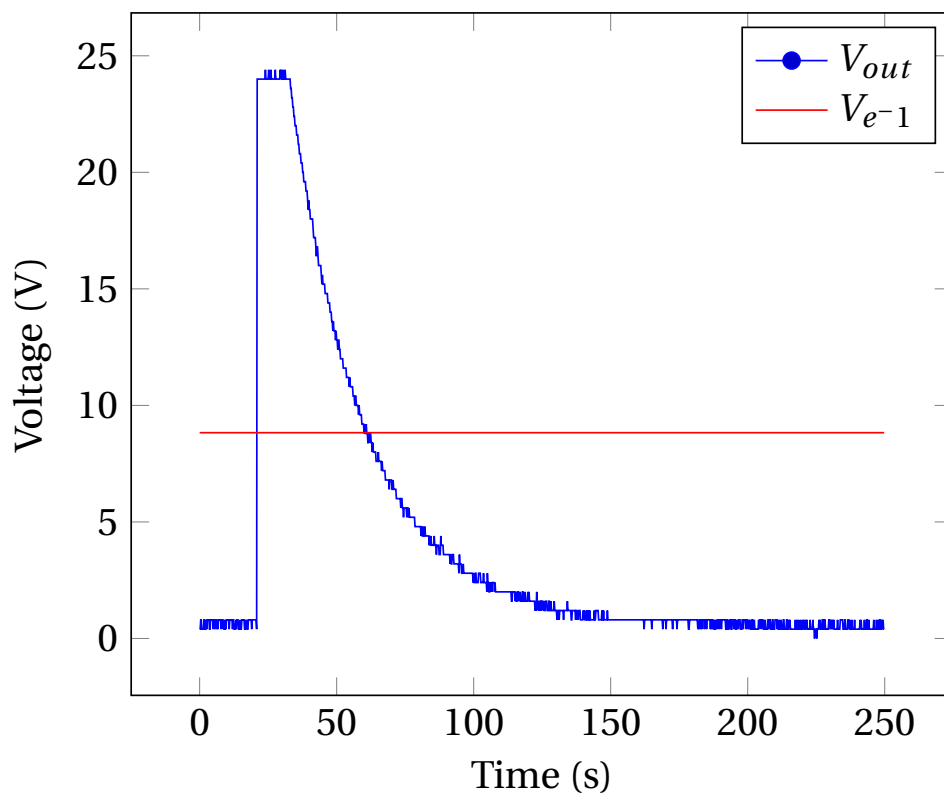


Figure B.5: The discharge curve of the capacitors.

From figure B.5 it can be seen that the capacitors are loaded at $t = 21.0\text{s}$. The curve crosses the red line (which indicates the point that τ is 1) at $t \approx 62\text{s}$. This means $\tau \approx 41\text{s}$, which means that the capacitors will be discharged after 205 seconds, which fits nicely in the 2.5 minute limit.

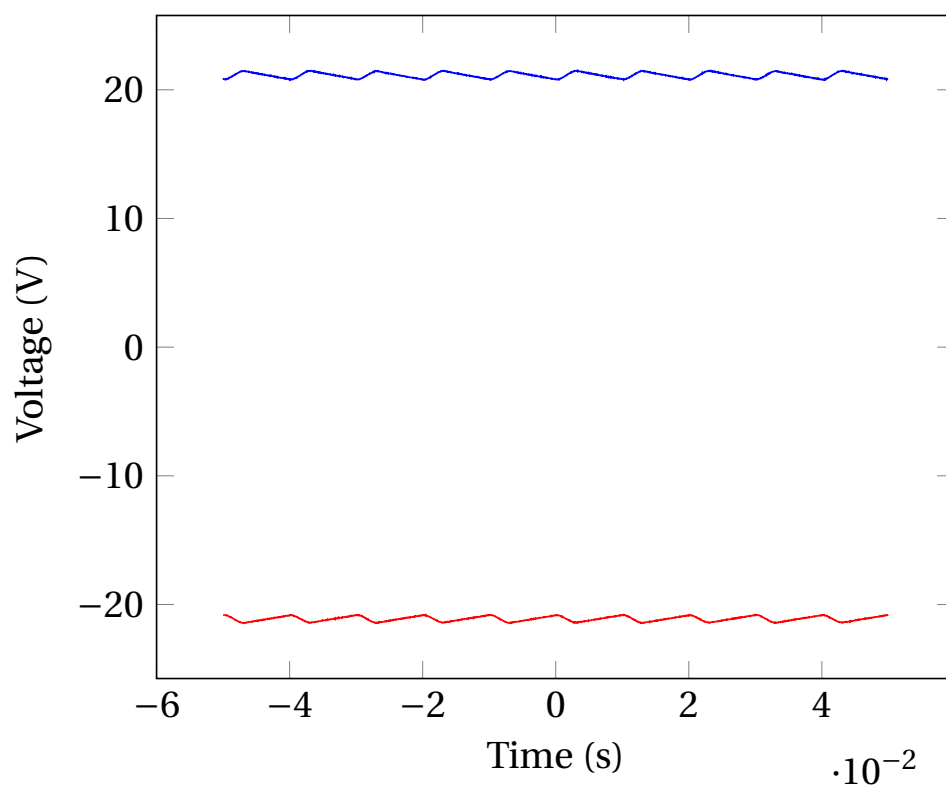


Figure B.6: The output voltage waveform with the power supply loaded with a $38\ \Omega$ resistor.